Ex 1

16/7/26

Study of various network commands used in Linux and Windows

Aim :-

To study the various network commands used in linux and windows.

Basic networking commands :-

1. arp

Show the IP address of our computer along with IP address and MAC address of our router.

2. hostname

Simplest of all TCP/IP commands. It simply displays the name of our computer.

3. ip config

Displays detailed configuration information about TCP/ IP connection including router, gateway and so on.

nbstat

Helps solve problems with NetBIOS name resolution.

Ex 1
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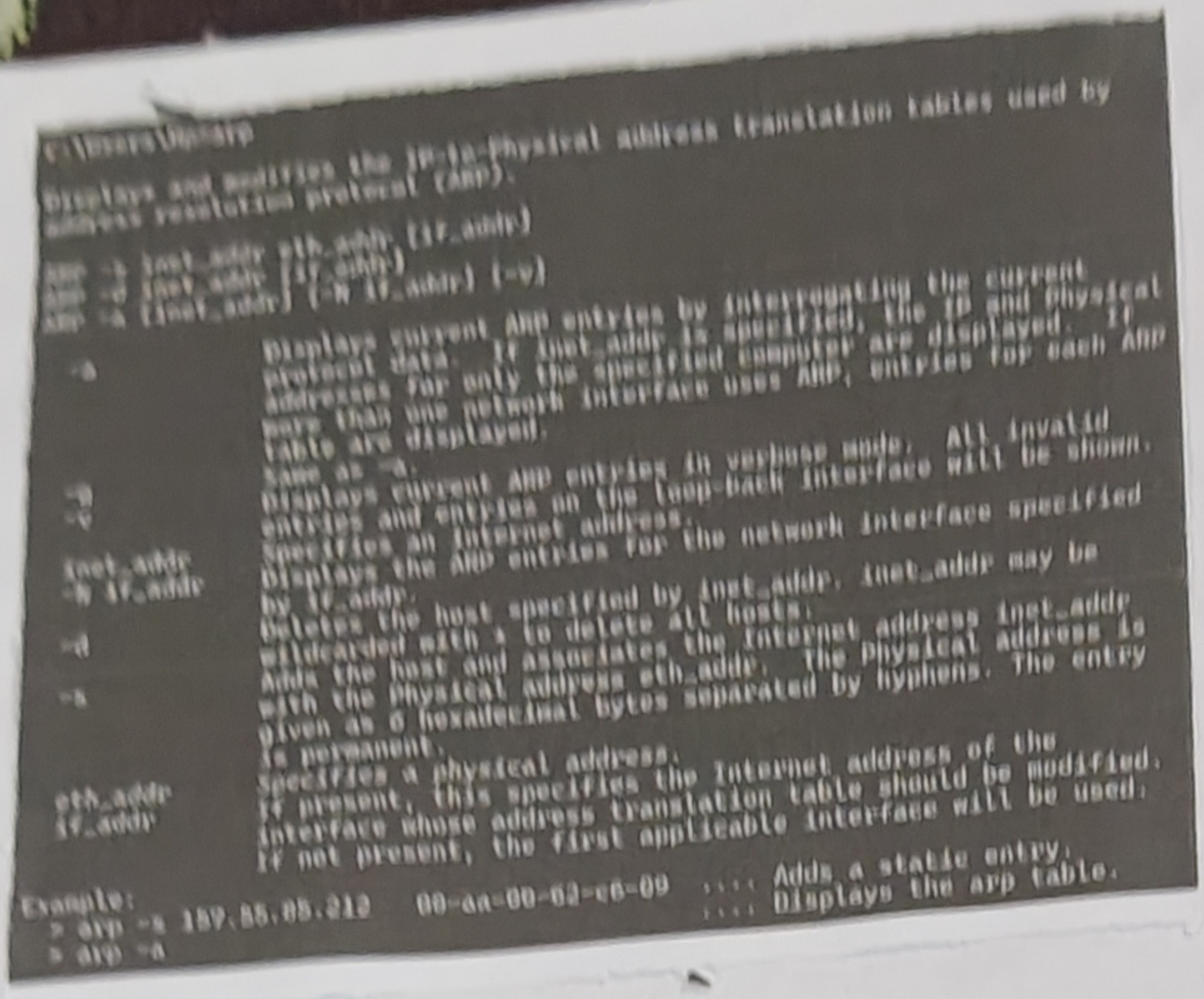
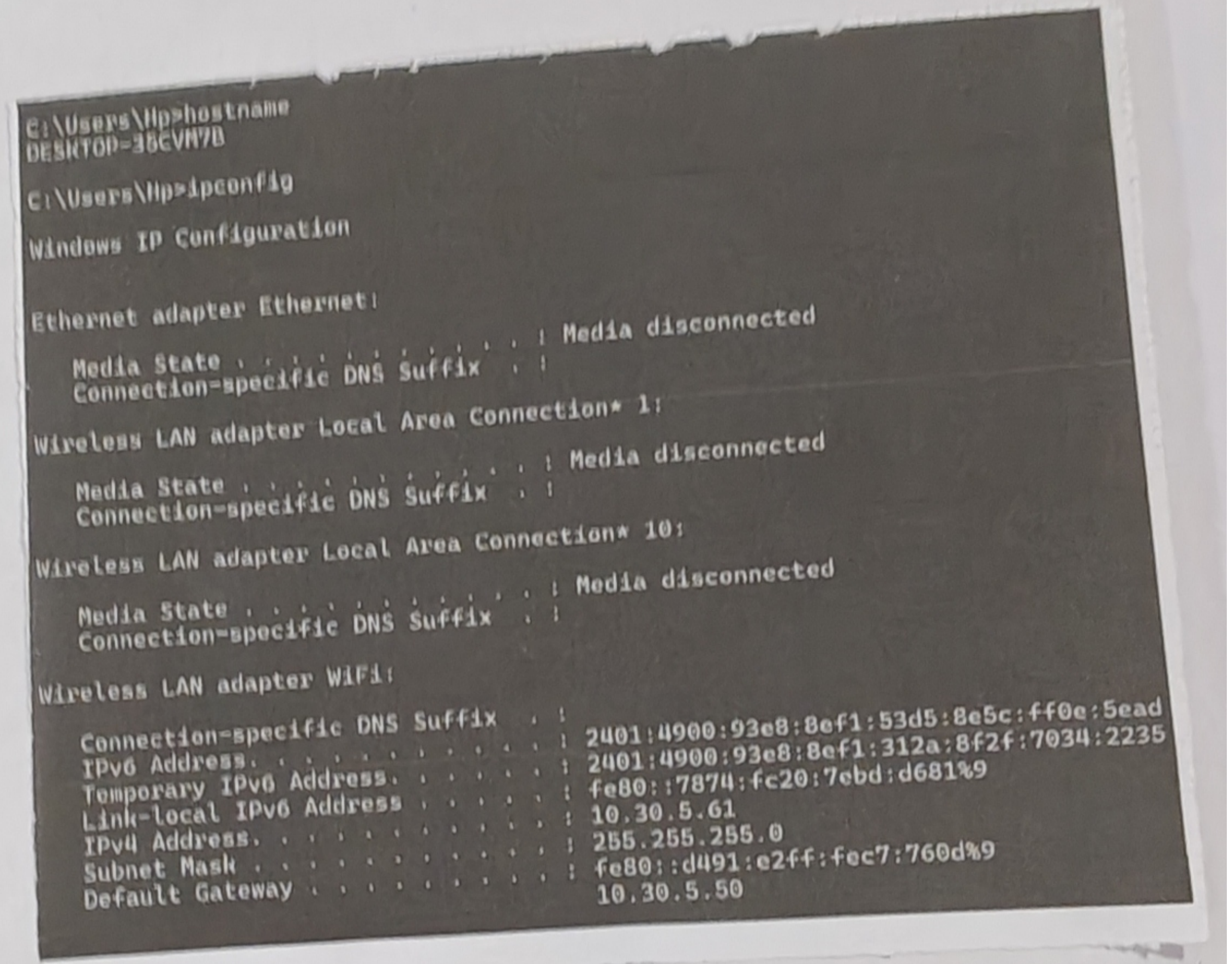
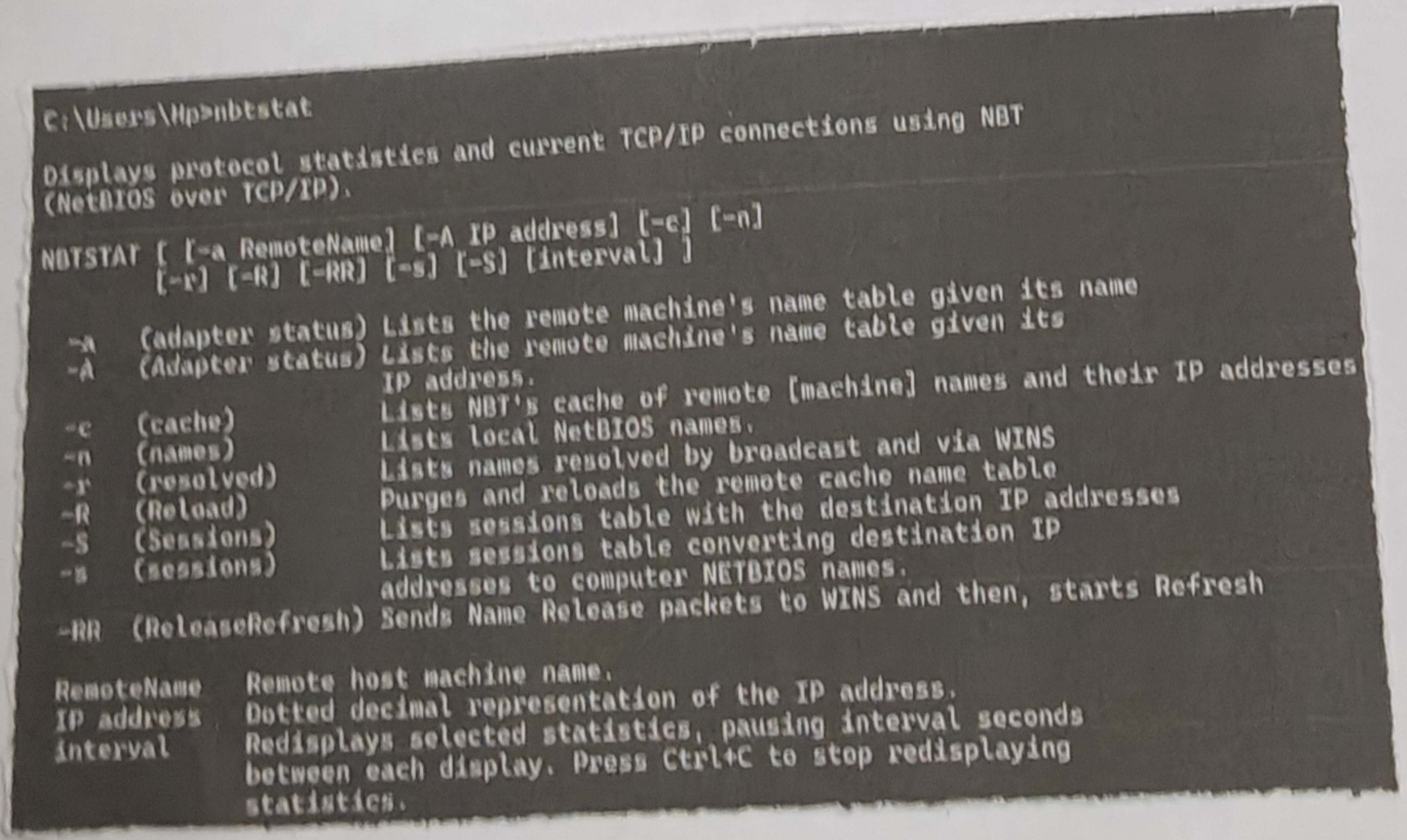


```
C:\Users\Hp>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```



```
C:\Users\Hp>hostname
DESKTOP-35CVM7B

C:\Users\Hp>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:93e8:8ef1:53d5:8e5c:ff0e:5ead
   Temporary IPv6 Address. . . . . . : 2401:4900:93e8:8ef1:312a:8f2f:7034:2235
   Link-local IPv6 Address . . . . . : fe80::7874:fc20:7ebd:d681%9
   IPv4 Address. . . . . . . . . . . : 10.30.5.61
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::d491:e2ff:fec7:760d%9
                                       10.30.5.50
```



```
C:\Users\Hp>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

5. nslookup

used to perform DNS lookups in linux. Displays the DNS details such as IP address of particular computer.

6. pathping

Unique to windows and Basically a combination of ping and tracert commands.

7. ping

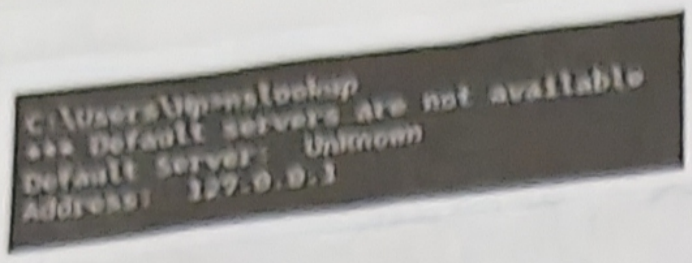
Best way to test connectivity Betwee two nodes. It uses ICMP to communic to other devices.

8. Route

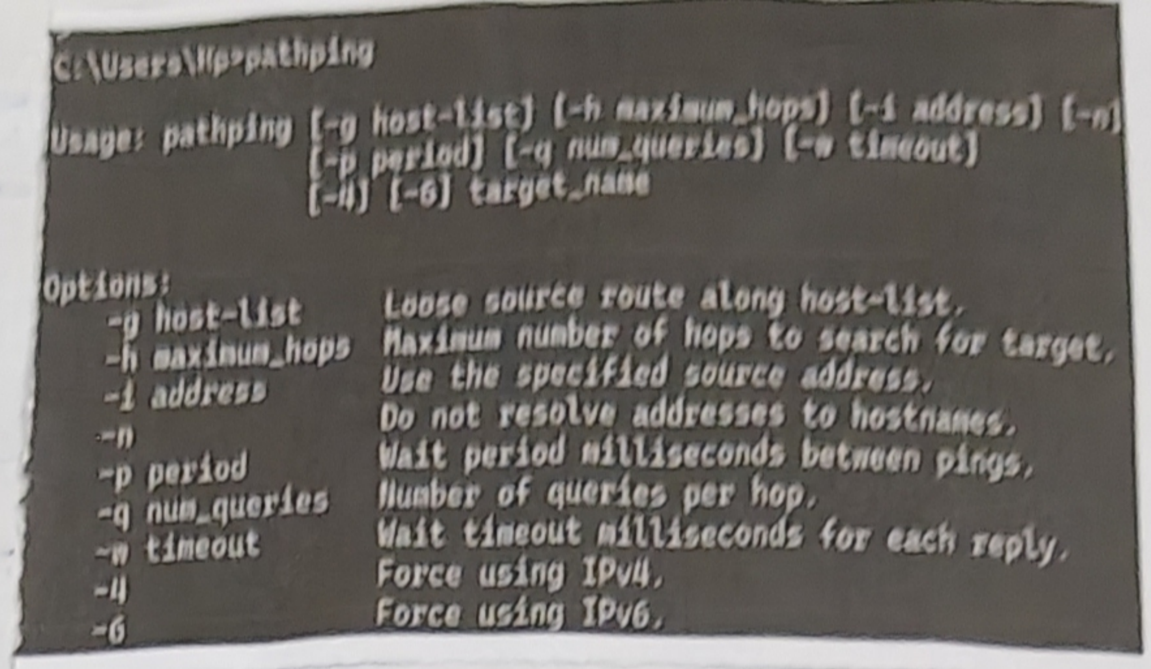
used to show/manipulate IP routing table. It is primarily used to setup static routes to specific host

9. netstat

Displays variety of statistics about computers active TCP/IP connections.

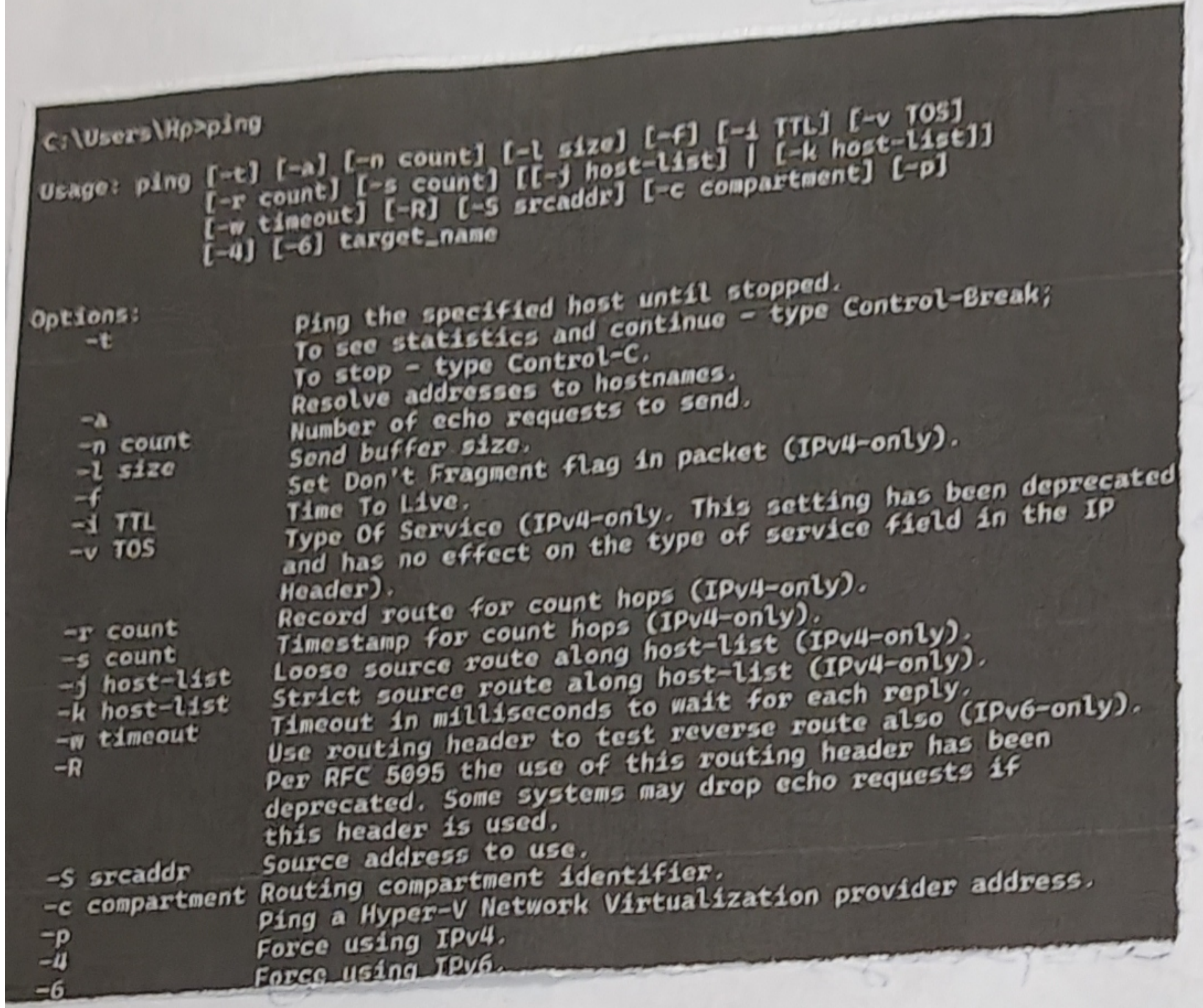


```
C:\Users\Hp>nslookup
*** Default servers are not available
Default Server:  UnKnown
Address:  127.0.0.1
```



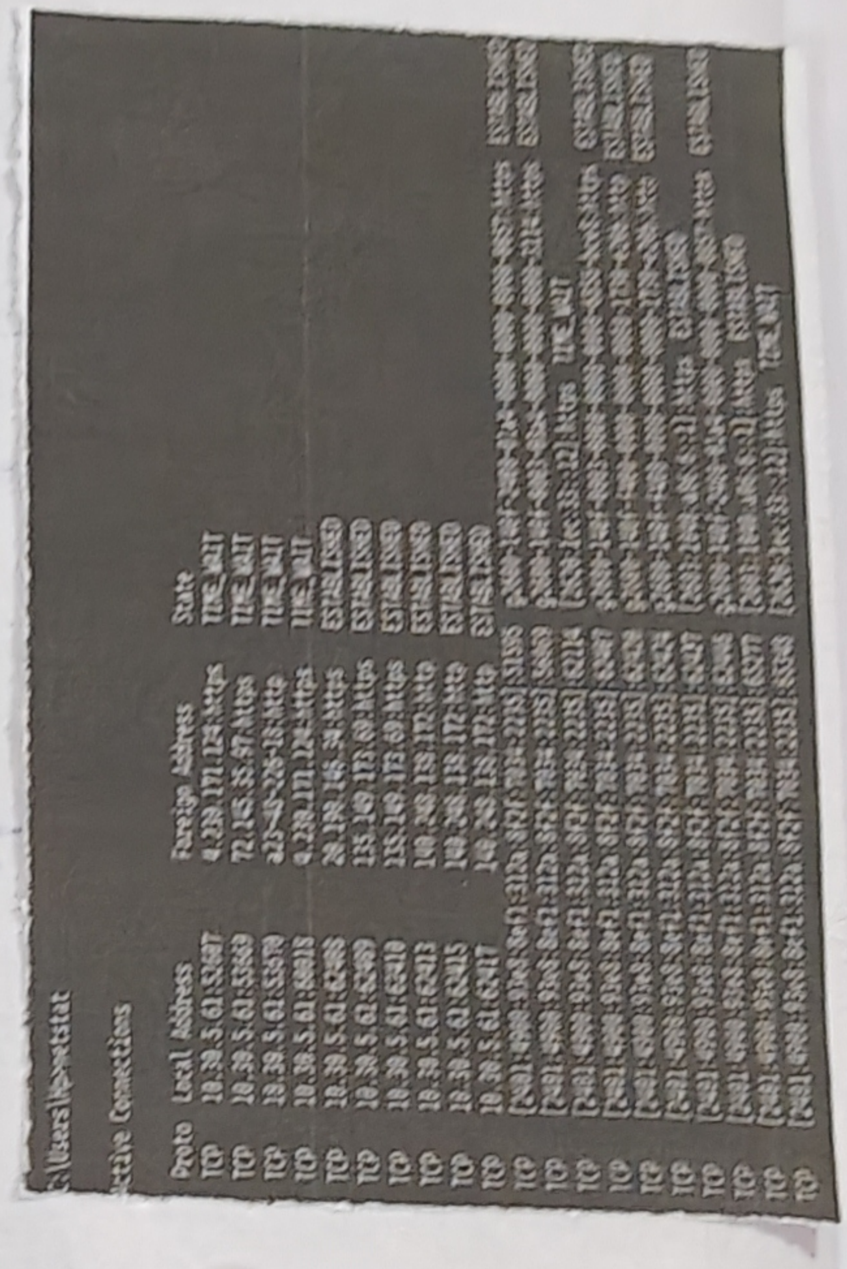
```
C:\Users\Hp>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

5.



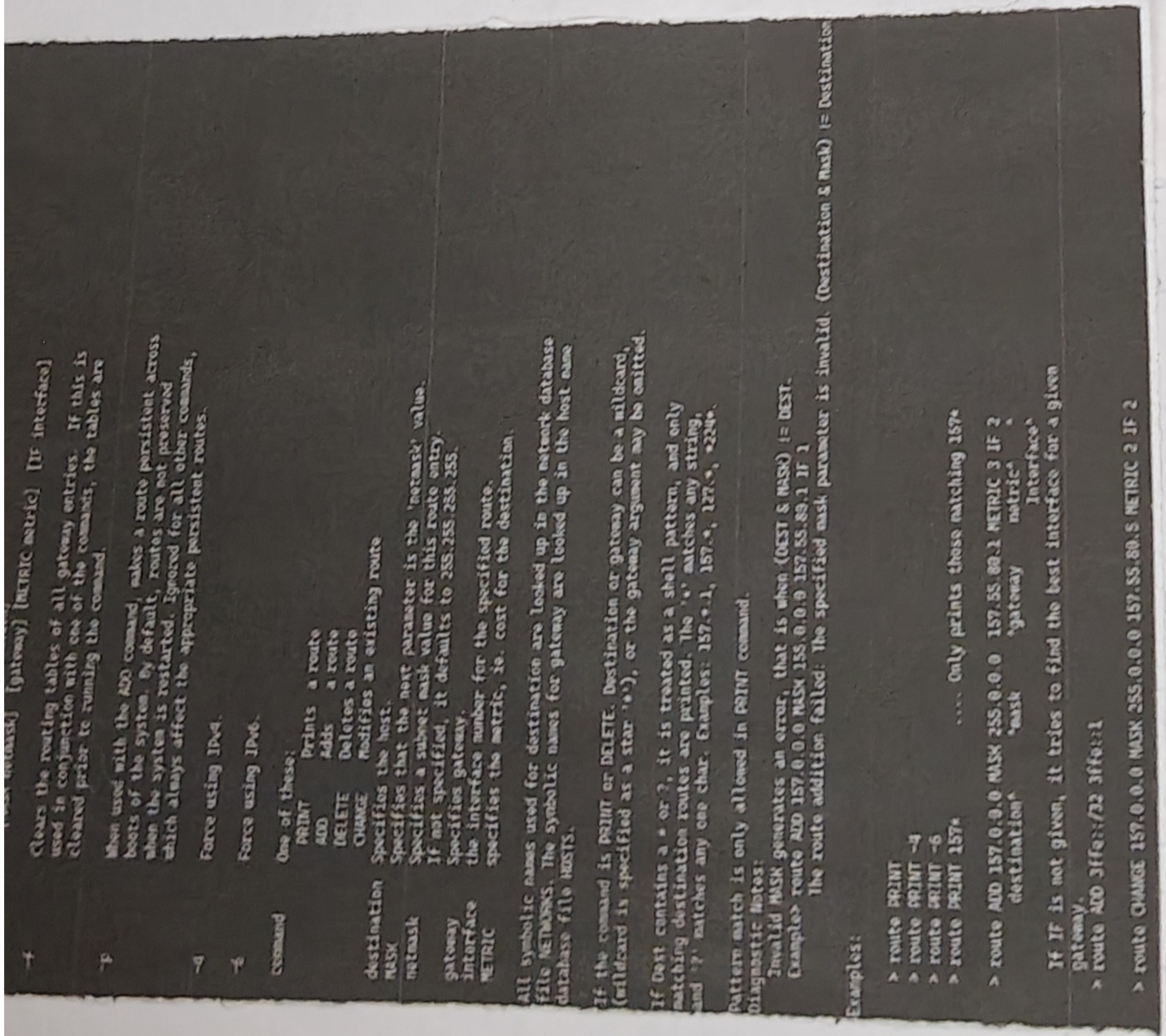
```
C:\Users\Hp>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

6-



```
C:\Users\Hp>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.30.5.61:52607       4.239.171.124:https    TIME_WAIT
  TCP    10.30.5.61:53669       72.145.35.97:https     TIME_WAIT
  TCP    10.30.5.61:53670       a23-48-226-18:http     TIME_WAIT
  TCP    10.30.5.61:60018       4.239.171.124:https    TIME_WAIT
  TCP    10.30.5.61:62408       20.198.146.34:https    ESTABLISHED
  TCP    10.30.5.61:62409       135.149.173.69:https   ESTABLISHED
  TCP    10.30.5.61:62410       135.149.173.69:https   ESTABLISHED
  TCP    10.30.5.61:62413       140.248.138.172:http   ESTABLISHED
  TCP    10.30.5.61:62415       140.248.138.172:http   ESTABLISHED
  TCP    10.30.5.61:62417       140.248.138.172:http   ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:51595  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:56069  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62114  [2620:1ec:33::22]:https  TIME_WAIT
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62407  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62420  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62424  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62427  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:62466  [illegible]  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:65277  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2401:4900:93e8:8ef1:312a:8f2f:7034:2235]:65340  [2620:1ec:33::22]:https  TIME_WAIT
```

7-



```
                  netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
               If not specified, it defaults to 255.255.255.255.
  gateway      Specifies gateway.
  interface    the interface number for the specified route.
  METRIC       specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*          .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
             destination^      ^mask      ^gateway     metric^    ^
                                                         Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2
```

8

Some important linux networking commands

1- ip

Basic command needed from setting up new systems and assigning IPs to troubleshooting existing systems.

2- ipconfig

Tool belt for configuring and troubleshooting networks.

3- mtr

Serves as a network diagnostic and troubleshooting tool. combines functionality of ping and traceroute commands.

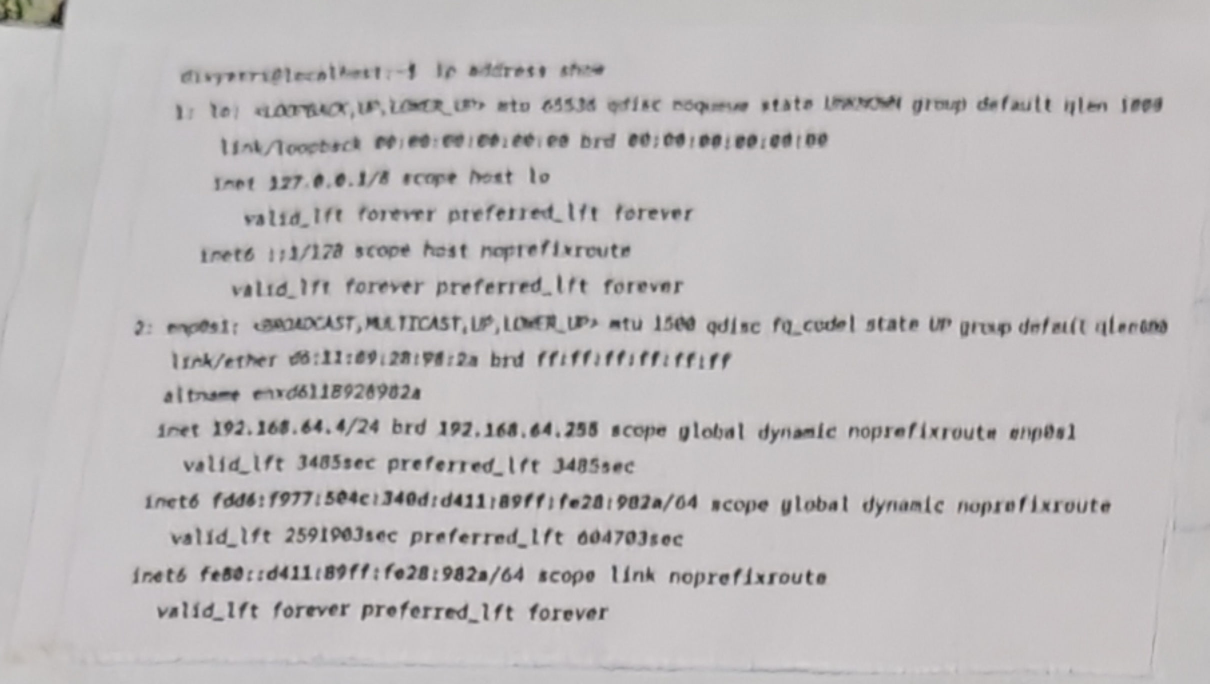
4. tcpdump

Designed for capturing and displaying packets.

5- ping

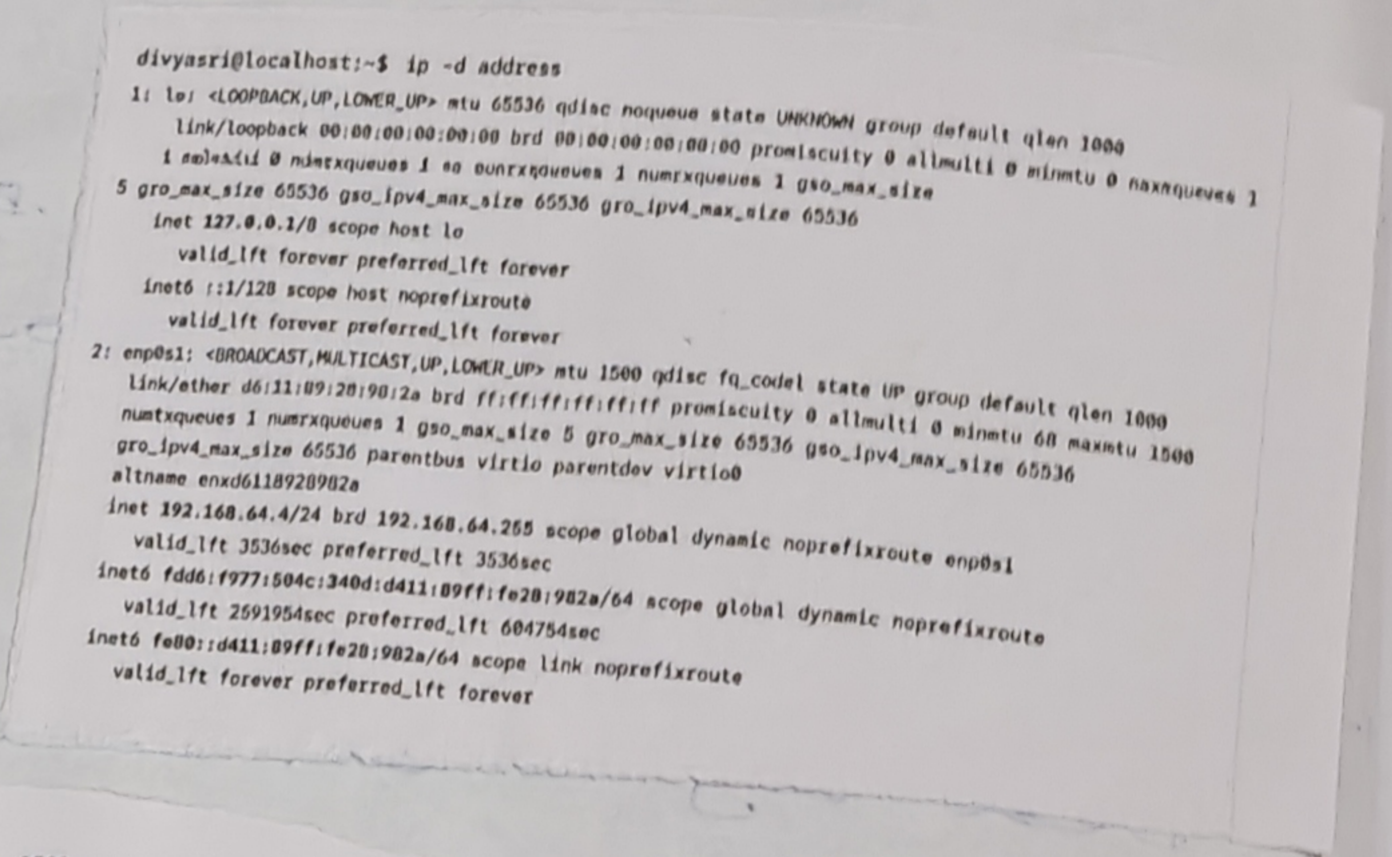
Tool that verifies IP level connectivi to another TCP/IP computer by sendi Internet Control Message Protocol.

1-



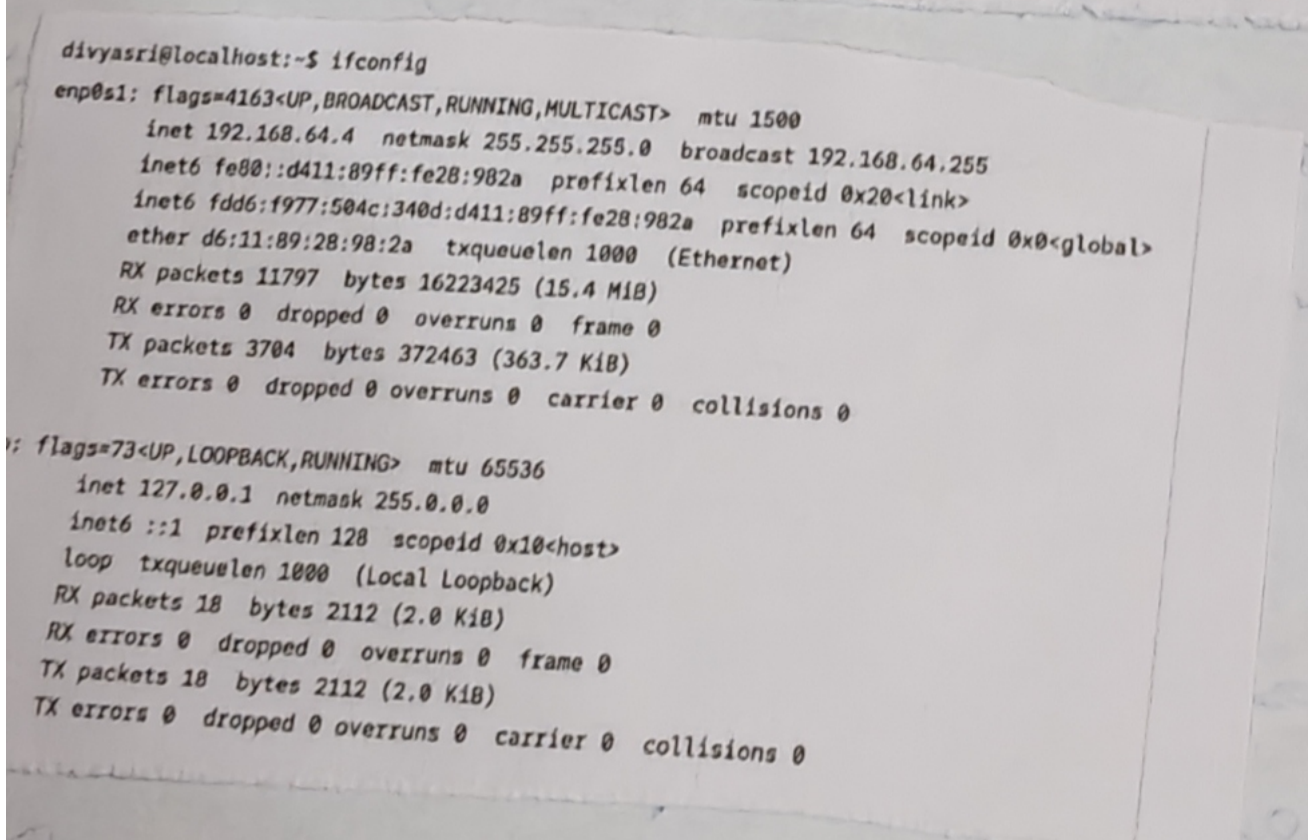
```
divyasri@localhost:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen1000
    link/ether d6:11:89:28:98:2a brd ff:ff:ff:ff:ff:ff
    altname enxd6118928982a
    inet 192.168.64.4/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
       valid_lft 3485sec preferred_lft 3485sec
    inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a/64 scope global dynamic noprefixroute
       valid_lft 2591903sec preferred_lft 604703sec
    inet6 fe80::d411:89ff:fe28:982a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

2-



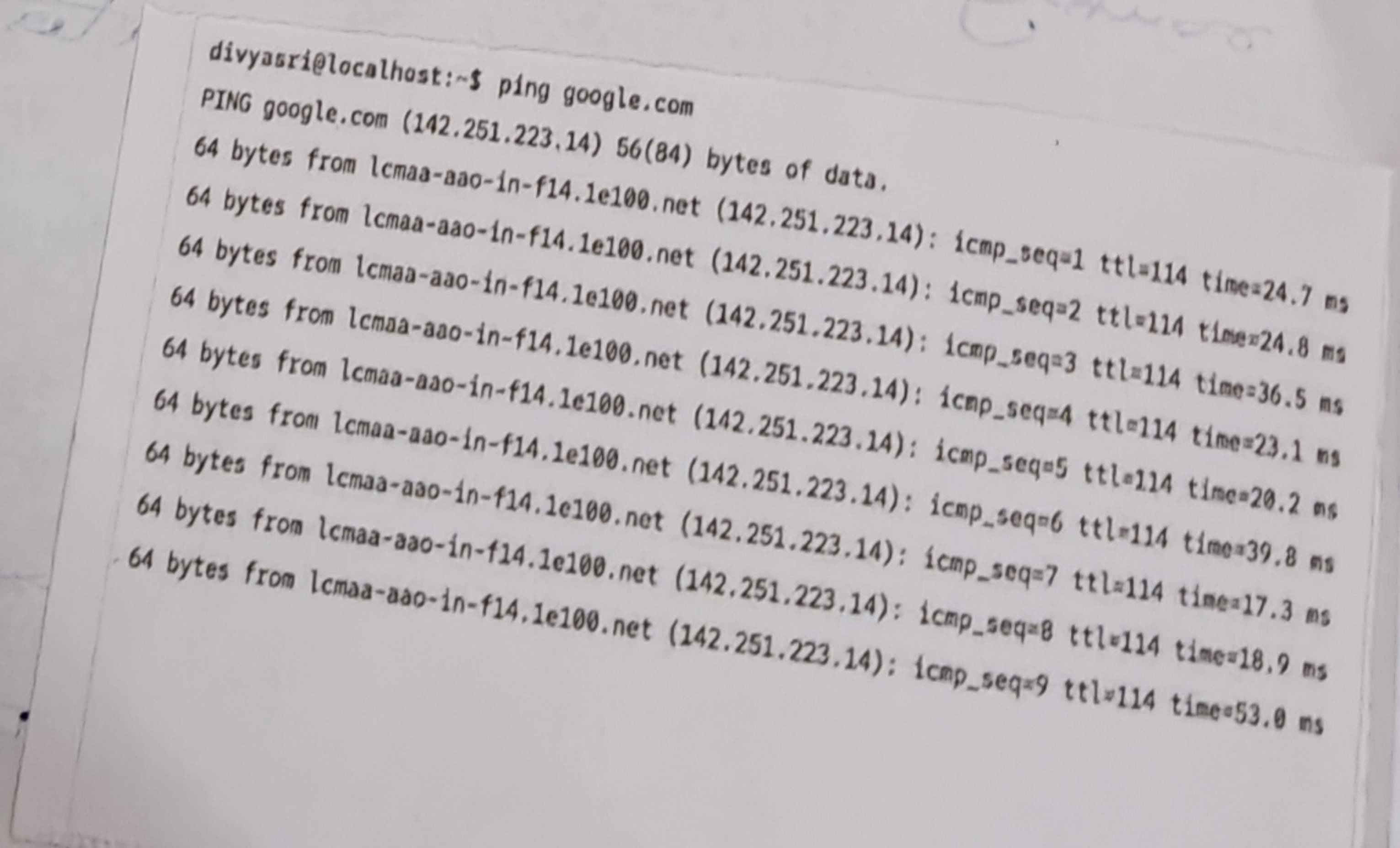
```
divyasri@localhost:~$ ip -d address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmqueues 1
    1 amlesli 0 numtxqueues 1 ao ounrxqueues 1 numrxqueues 1 gso_max_size
5 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether d6:11:89:28:98:2a brd ff:ff:ff:ff:ff:ff promiscuity 0 allmulti 0 minmtu 68 maxmtu 1500
    numtxqueues 1 numrxqueues 1 gso_max_size 5 gro_max_size 65536 gso_ipv4_max_size 65536
    gro_ipv4_max_size 65536 parentbus virtio parentdev virtio0
    altname enxd6118928982a
    inet 192.168.64.4/24 brd 192.168.64.255 scope global dynamic noprefixroute enp0s1
       valid_lft 3536sec preferred_lft 3536sec
    inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a/64 scope global dynamic noprefixroute
       valid_lft 2591954sec preferred_lft 604754sec
    inet6 fe80::d411:89ff:fe28:982a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

3-



```
divyasri@localhost:~$ ifconfig
enp0s1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.64.4  netmask 255.255.255.0  broadcast 192.168.64.255
        inet6 fe80::d411:89ff:fe28:982a  prefixlen 64  scopeid 0x20<link>
        inet6 fdd6:f977:504c:340d:d411:89ff:fe28:982a  prefixlen 64  scopeid 0x0<global>
        ether d6:11:89:28:98:2a  txqueuelen 1000  (Ethernet)
        RX packets 11797  bytes 16223425 (15.4 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3704  bytes 372463 (363.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 18  bytes 2112 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 18  bytes 2112 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

4.



```
divyasri@localhost:~$ ping google.com
PING google.com (142.251.223.14) 56(84) bytes of data.
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=1 ttl=114 time=24.7 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=2 ttl=114 time=24.8 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=3 ttl=114 time=36.5 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=4 ttl=114 time=23.1 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=5 ttl=114 time=20.2 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=6 ttl=114 time=39.8 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=7 ttl=114 time=17.3 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=8 ttl=114 time=18.9 ms
64 bytes from lcmaa-aao-in-f14.1e100.net (142.251.223.14): icmp_seq=9 ttl=114 time=53.0 ms
```

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Result :-

Thus studied the various network commands used in linux and windows and implemented it.